AGA KHAN UNIVERSITY EXAMINATION BOARD

SECONDARY SCHOOL CERTIFICATE

CLASS X EXAMINATION

MAY 2011

General Mathematics Paper I

Time allowed: 40 minutes Marks 30

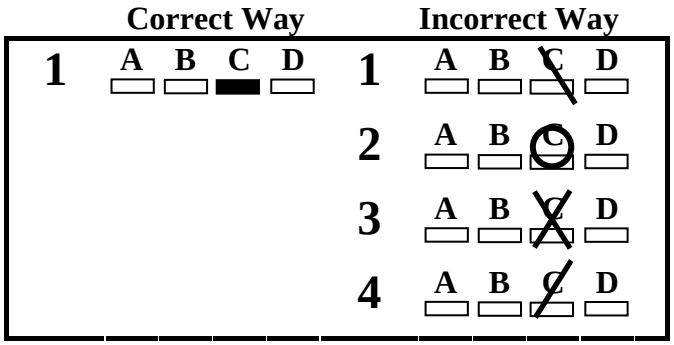
INSTRUCTIONS

1. Read each question carefully.

2. Answer the questions on the separate answer sheet provided. DO NOT write your answers on the question paper.

3. There are 100 answer numbers on the answer sheet. Use answer numbers 1 to 30 only.

4. In each question there are four choices A, B, C, D. Choose ONE. On the answer grid black out the box for your choice with a pencil as shown below.



5. If you want to change your answer, ERASE the first answer completely with a rubber, before blacking out a new box.

6. DO NOT write anything in the answer grid. The computer only records what is in the boxes.

7. You may use a simple calculator if you wish.

1. If $3x - 6 = 24$, then x is equal to

   A. 6
   B. 10
   C. 15
   D. 25

2. If $x = 3$ and $y = 1$ then $x^2 - y^2$ is equal to

   A. 4
   B. 5
   C. 7
   D. 8

3. Which one of the following is a polynomial?

   A. $\sqrt{x^3}$
   B. $x^2 + \frac{1}{x^2}$
   C. $x^2 + \frac{1}{\sqrt{2}}$
   D. $\sqrt{x^2 + 1}$

4. If $x + 1$ is a factor of $x^3 - m$, then m is equal to

   A. –1
   B. 0
   C. 1
   D. 2

5. The factorization of $n^3 + n$ is

   A. $n\left(n^2 + 1\right)$
   B. $n(n+1)(n-1)$
   C. $n^2(n+1)$
   D. $n\left(n^2 + n\right)$

6. If $x^2 - 5x - 36 = (x+4)(x+P)$, then the value of P is

   A. 9
   B. 6
   C. – 6
   D. – 9

7. On simplification of $\frac{x^2 - y^2}{x - y} \div (x + y)$, we get

A. 0
B. 1
C. $x + y$
D. $(x + y)^2$

8. The Highest Common Factor (H.C.F) of $3a$, $6a^2$ and $9a^3$ is

A. $3a$
B. $6a$
C. $3a^2$
D. $6a^2$

9. To make $x^2 + \frac{1}{4}$ a perfect square, we should add

A. 1
B. $\frac{x}{2}$
C. $-x$
D. $2x$

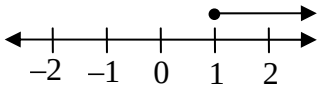
10. $\frac{1}{x^2} - \frac{1}{x}$ is equal to

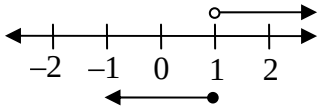
A. $\frac{1-x}{x^2}$
B. $\frac{x-1}{x^2}$
C. $\frac{1-x}{x}$
D. $\frac{x-1}{x}$

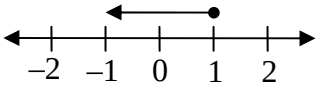
11. If $|x - d| = 0$, then value of x is equal to

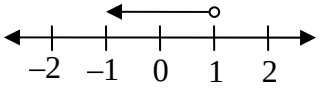
A. $-d$
B. d
C. d and $-d$
D. 0 and d

12. If x is a real number, then the solution of $\frac{3}{2}x-\frac{3}{2}>0$ can be represented on a number line as

A. 

B. 

C. 

D. 

13. The solution set of $x^2=5$ is

A. $\{(0,\sqrt{5})\}$
B. $\{(0,-\sqrt{5})\}$
C. $\{(\sqrt{5},-\sqrt{5})\}$
D. $\{(0,\pm\sqrt{5})\}$

14. The solution set of $\sqrt{x}=x$ is

A. $\{0\}$
B. $\{1\}$
C. $\{-1\}$
D. $\{0,1\}$

15. The matrix $\begin{bmatrix}1\\1\\1\end{bmatrix}$ is a

A. column matrix.
B. scalar matrix.
C. row matrix.
D. unit matrix.

16. The multiplicative identity of every 2×2 matrix is

A. $\begin{bmatrix}1&0\\1&0\end{bmatrix}$ B. $\begin{bmatrix}0&1\\1&0\end{bmatrix}$ C. $\begin{bmatrix}0&0\\0&0\end{bmatrix}$ D. $\begin{bmatrix}1&0\\0&1\end{bmatrix}$

17. Which one of the following is a skew-symmetric matrix?

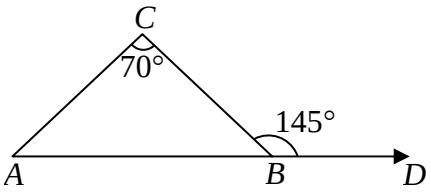
A. $\begin{bmatrix}0&2\\2&0\end{bmatrix}$ B. $\begin{bmatrix}0&2\\-2&0\end{bmatrix}$ C. $\begin{bmatrix}2&0\\0&2\end{bmatrix}$ D. $\begin{bmatrix}2&2\\0&0\end{bmatrix}$

18. If $Y - \begin{bmatrix} 4 \\ 7 \end{bmatrix} = \begin{bmatrix} 2 \\ 5 \end{bmatrix}$, then Y is equal to

A. $\begin{bmatrix} 2 \\ 2 \end{bmatrix}$ B. $\begin{bmatrix} 6 \\ 2 \end{bmatrix}$ C. $\begin{bmatrix} 2 \\ 12 \end{bmatrix}$ D. $\begin{bmatrix} 6 \\ 12 \end{bmatrix}$

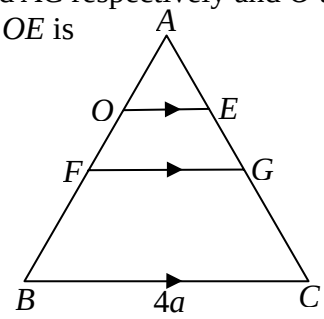
19. The measure of $\angle CAB$ in the given diagram is

A. $35°$.
B. $75°$.
C. $110°$.
D. $145°$.



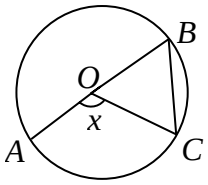
20. In the given triangle, if F and G are the mid points of AB and AC respectively and O and E are the mid point of AF and AG respectively, then the length of OE is

A. $2a$
B. $\frac{2}{3}a$
C. a
D. $\frac{a}{2}$



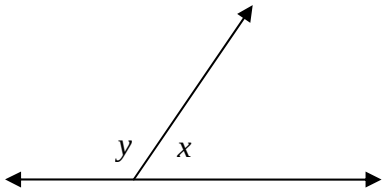
21. In the given figure, O is the centre of the circle and $OC = BC$, then the value of x is

A. $120°$.
B. $110°$.
C. $90°$.
D. $60°$.



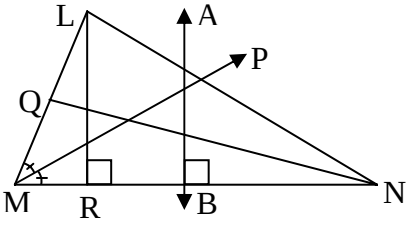
22. In the given figure if $y = 3x$, then y is equal to

A. $45°$.
B. $90°$.
C. $135°$.
D. $145°$.



23. In the given figure one of the altitudes is

A. $\overleftrightarrow{AB}$.
B. $\overrightarrow{MP}$.
C. $\overline{NQ}$.
D. $\overline{LR}$.

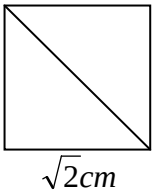


24. If the circumference and the area of a circle are a cm and $a\,\text{cm}^2$ respectively, then the radius of the circle is equal to

A. $\frac{1}{4}$ cm.
B. $\frac{1}{2}$ cm.
C. 1 cm.
D. 2 cm.

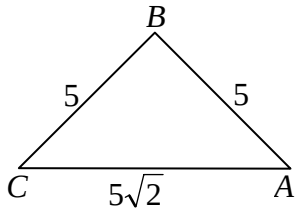
25. The length of the diagonal of a square whose measure of a side is $\sqrt{2}$ cm is

A. 16 cm.
B. 8 cm.
C. 4 cm.
D. 2 cm.



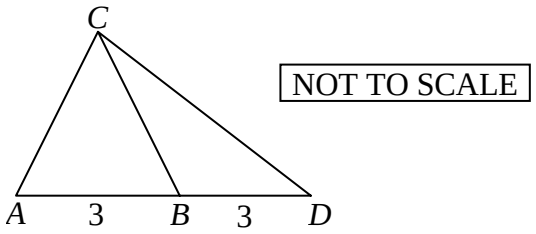
26. In the following right angled triangle, the hypotenuse and right angle are

A. $\overline{AB}$ and $\angle B$.
B. $\overline{AC}$ and $\angle A$.
C. $\overline{AC}$ and $\angle B$.
D. $\overline{BC}$ and $\angle A$.



27. Consider the following figure. If the area of ΔABC is equal to $6cm^2$, then the area of ΔBDC is

A. $6cm^2$
B. $9cm^2$
C. $12cm^2$
D. $18cm^2$



28. If the distance between AB, BC and AC is $\sqrt{2}$, $\sqrt{2}$, $2\sqrt{2}$ respectively, then which of the following is true

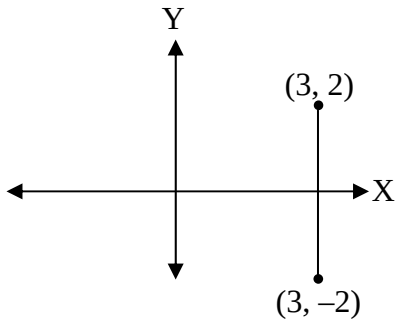
I. A, B and C are collinear points
II. A, B and C are non-collinear points
III. A, B and C form a triangle

A. I only
B. II only
C. III only
D. I and III only

29. The distance between points (0, 0) and (–1, –1) is

A. $-\sqrt{2}$ unit.
B. -1 unit.
C. 1 unit.
D. $\sqrt{2}$ unit.

30. The distance between the points shown in the figure is



A. 4 unit.
B. 6 unit.
C. $4\sqrt{3}$ unit.
D. $6\sqrt{2}$ unit.

END OF PAPER

Please use this page for rough work